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The Editors-in-Chief
Science of Computer Programming

Dear Editors,

I submit the manuscript “Compare the requests, not their sizes: Resource side channels and sound token bounds for LLM workflows” for consideration as a research paper in *Science of Computer Programming*.

A program that calls a large language model can leak a secret without the secret reaching a prompt: if the secret decides which requests are sent, the bill and the network traffic can reveal it. The paper shows that no analysis that sees prompt text only through a size measure can be both sound and useful for such programs (a theorem mechanised in Coq); it replaces size comparison by a comparison of requests by content, with noninterference for every provider, tokenizer and observer of the transcript and a $\log_2 k$ bound on min-capacity; and it computes worst-case token bounds in bytes, converted through tokenizer contracts measured on fourteen tokenizers. Both analyses are implemented in a compiler for a small workflow language and evaluated on synthetic suites, one locally run model, and thirty workflows ported from public sources under a protocol fixed before porting.

The work fits the journal’s scope in programming languages, formal methods and program analysis, and it builds on lines of work the journal has published, among them static cost analysis of workflow languages (Ali, Lamo and Pun, SCP 225, 2023) and language-based information flow with declassification (Manzino and de Latorre, SCP 247, 2026). I am not aware of work in the journal on the analysis of LLM workflows.

The manuscript reports its negative results plainly. Two earlier relational rules of mine were unsound, and the paper explains why; and none of the thirty real workflows has a secret, so the side-channel analysis has so far been exercised only on synthetic workflows. The compiler, the Coq development, the benchmarks, the real-workflow corpus and the evaluation harness are publicly available, and three commands reproduce every number in the paper.

The manuscript is original, has not been published, and is not under consideration elsewhere. The use of generative AI in preparing the artifact and the manuscript is declared in the manuscript as the journal’s policy requires. I have no competing interests to declare.

Yours sincerely,

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